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Linux-palvelimet - ICI003AS2A-3021

Raportti 3
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Intro

The assignment was to install Apache and serve two named sites from this virtual machine. The names point at 127.0.0.1 through /etc/hosts, and the pages are plain HTTP on port 80.

Apache

I installed apache2. systemctl showed the service active (running) and enabled, so it starts again when the machine boots.

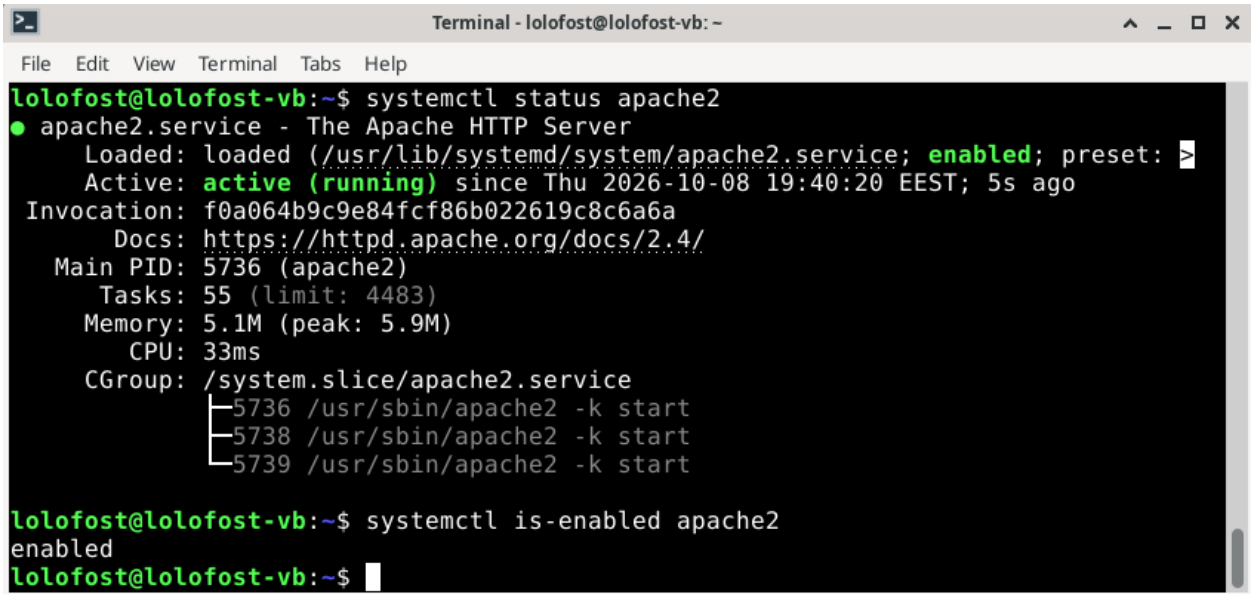


Image 1. Apache running.

I opened http://localhost in the browser. The Debian default page came up, which means Apache was answering.

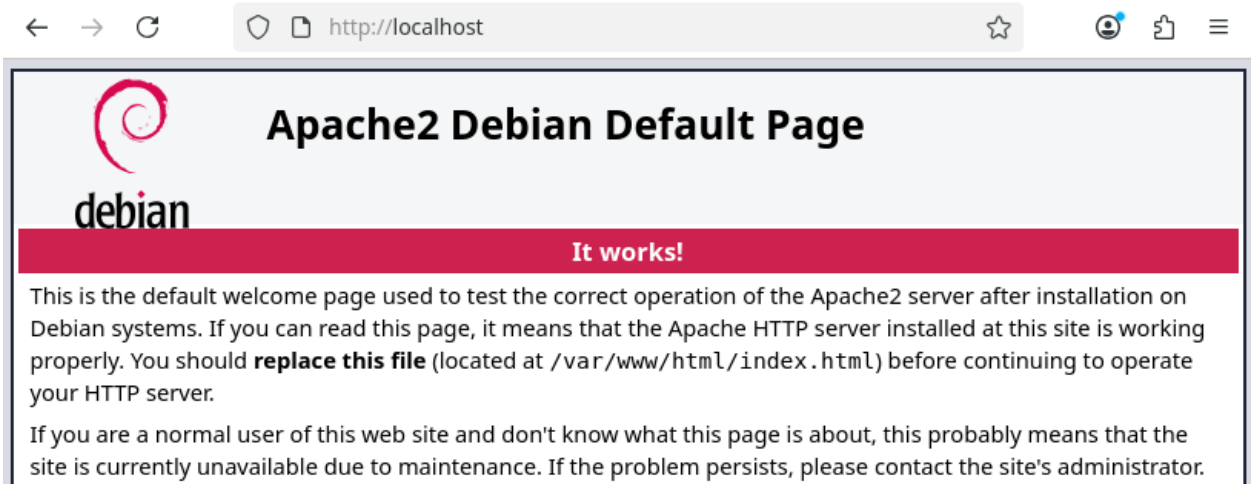


Image 2. The default page.

I replaced /var/www/html/index.html. echo prints the sentence, the pipe sends it to tee, and sudo runs tee as root so tee can write the file. tee also prints the sentence, which is why it showed up twice. curl localhost returned the same text.

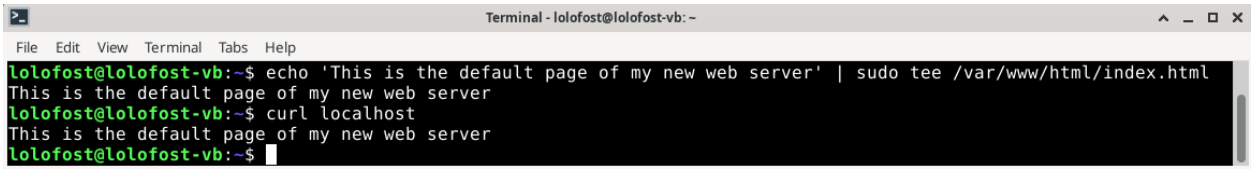


Image 3. The new default page.

sudo echo with > does not work for this file. The shell tries to open the file before sudo starts, and at that point I am still a normal user. The terminal says Permission denied. sudo nano is another way to edit the same file.

Names

I appended one line to /etc/hosts: 127.0.0.1 site1.com www.site1.com. Both names now mean this machine. tee -a adds the line. Without -a, tee would replace the whole file.



Image 4. The hosts entry for site1.com.

Firewall

I installed ufw. ufw allow OpenSSH failed, because this machine has no profile with that name. I allowed ports 22, 80 and 443 by number instead, then enabled the firewall. Port 22 is SSH, 80 is HTTP and 443 is HTTPS.

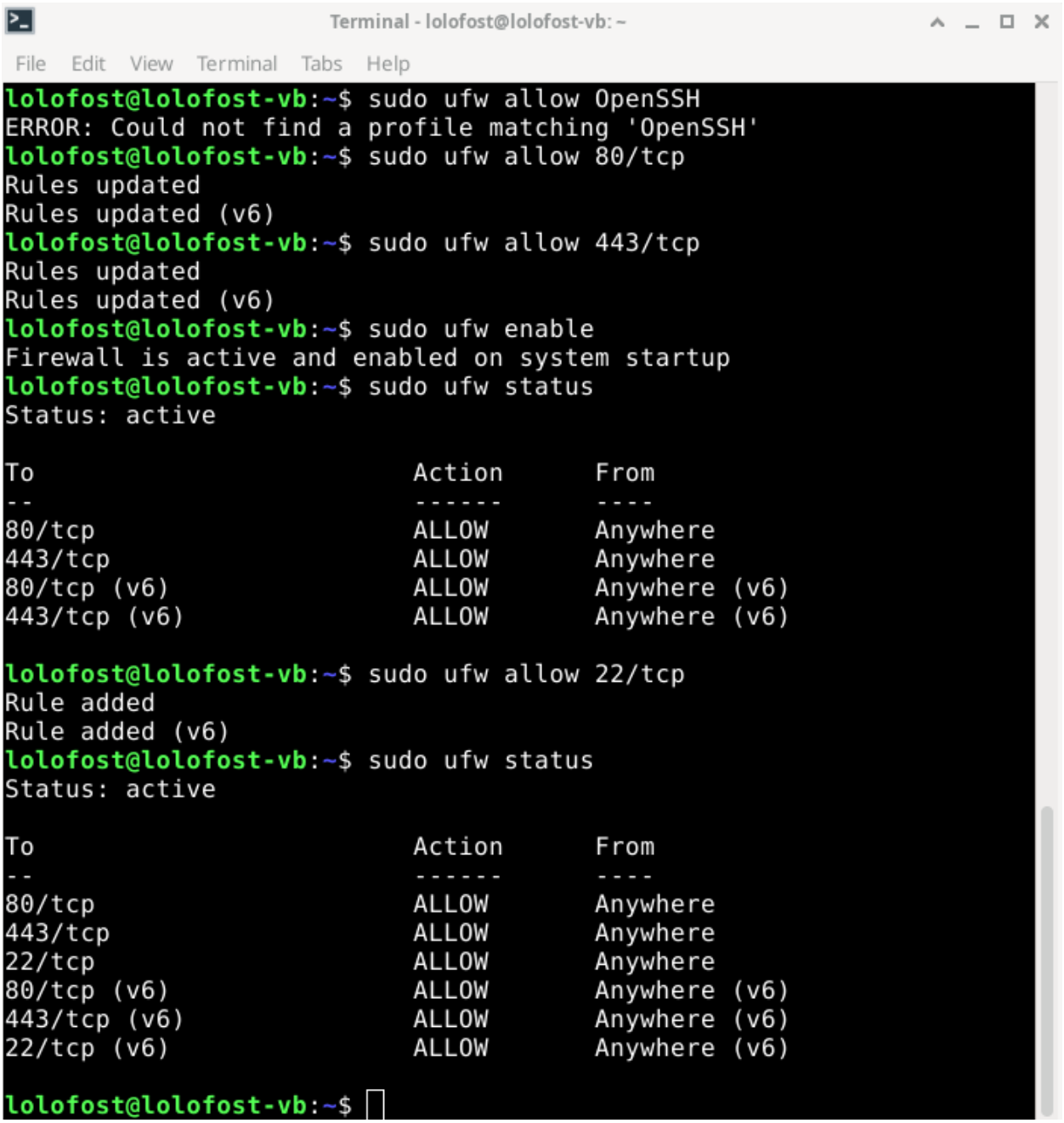


Image 5. The firewall after port 22 was added.

I removed the allow rule for port 80 and requested 127.0.0.1 and 10.0.2.15. Both still returned HTTP 200. The status at that point listed 22 and 443, not 80. On this virtual machine, taking port 80 out of the rules did not stop those local requests. I allowed port 80 again after the test.

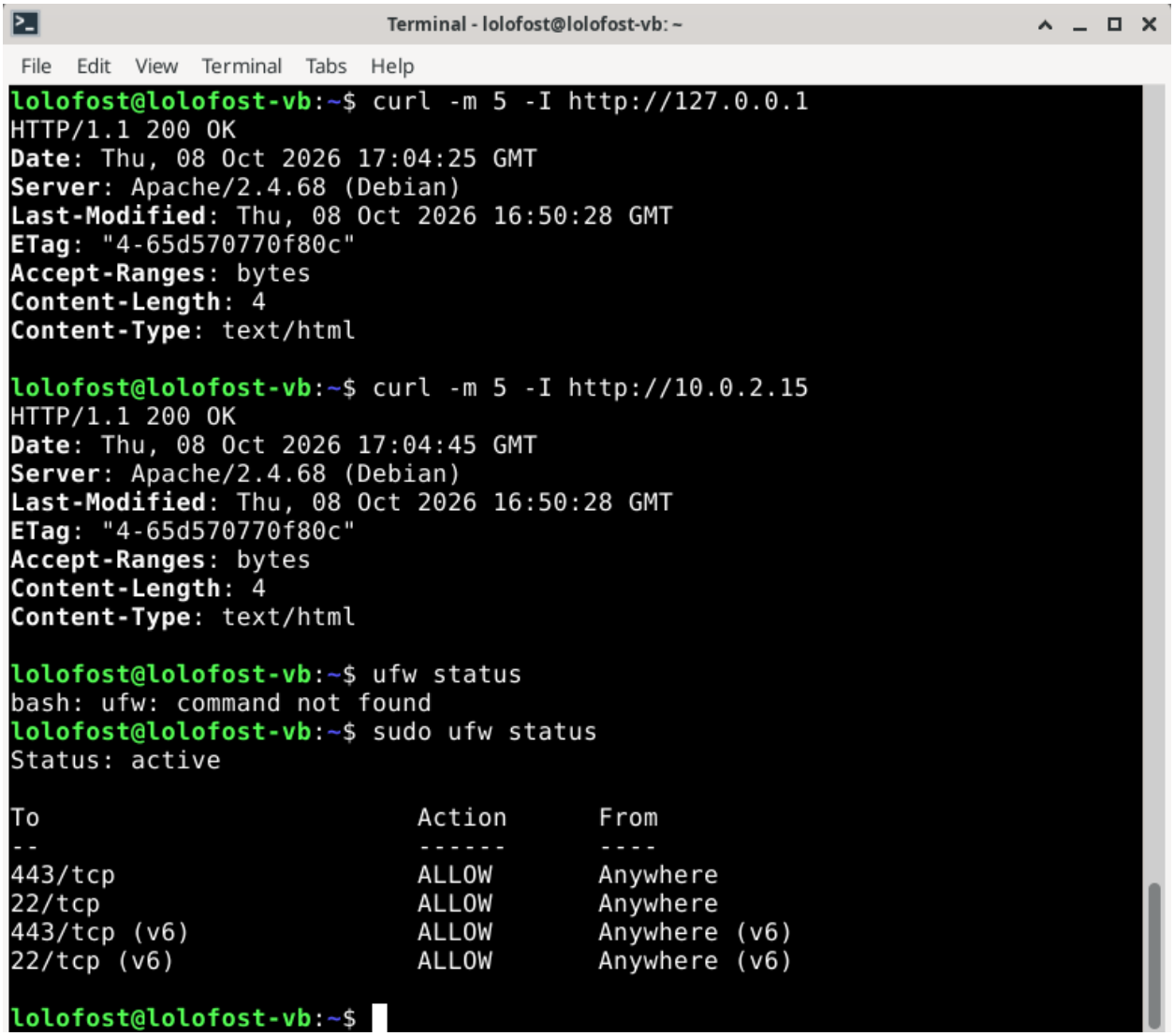


Image 6. Both local addresses still answered.

site1.com

The site files are in my home directory, in public-sites/index.html. Apache reads them as the user www-data, so www-data needs to walk the directories and read the file. /home and public-sites already allowed that. /home/lolofost did not: the listing was drwx------, so other users could not enter it. I added execute permission for others on that directory. index.html was already readable by others.

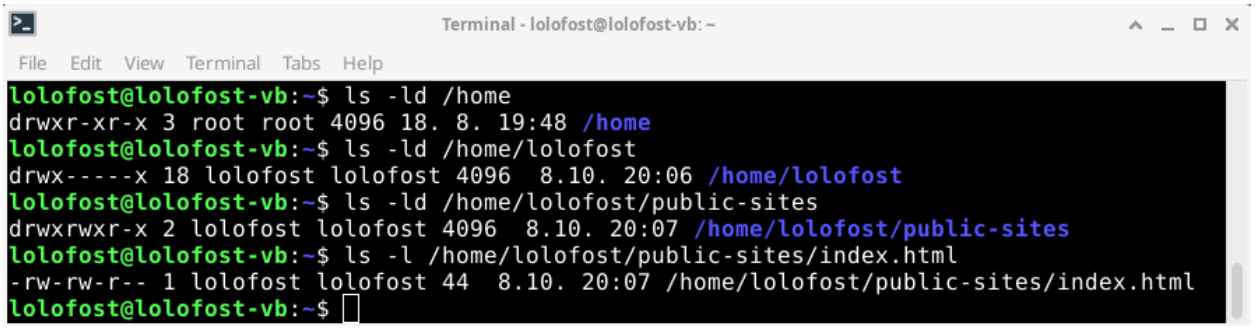


Image 7. The path before the home directory was opened.

The configuration listens on port 80. The name is site1.com, with www.site1.com as an alias, and the page directory is /home/lolofost/public-sites. The access and error logs for this site go to their own files under /var/log/apache2/.

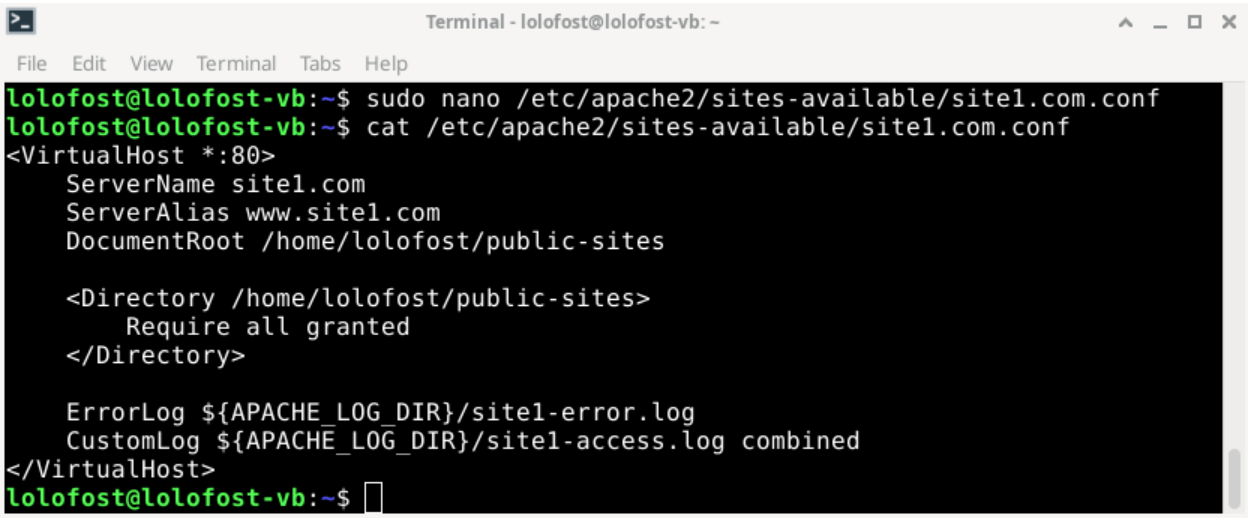


Image 8. The site1.com configuration.

a2ensite turned the site on by linking the file into sites-enabled. configtest printed Syntax OK. It also said Apache could not find a global ServerName and would use 127.0.1.1. That did not stop the site. curl on site1.com and on www.site1.com both returned the page.

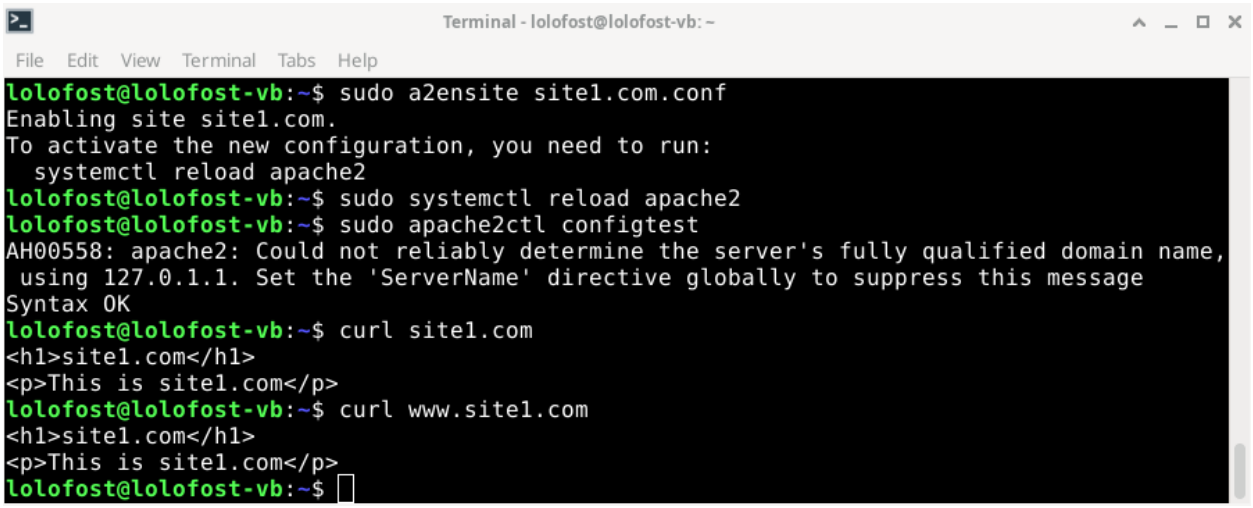


Image 9. Both names for the first site.

Logs

The site access log shows those two requests. Each one came from 127.0.0.1, asked for /, got status 200, and the client was curl.

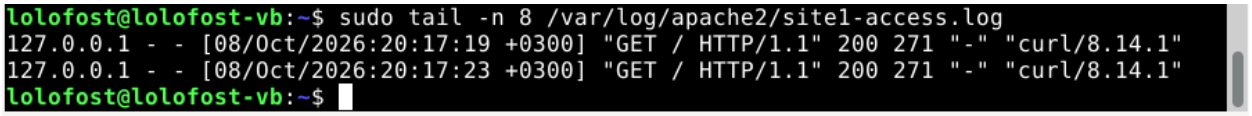


Image 10. The site1 access log.

I deleted the closing > from the VirtualHost line and reloaded. The reload failed. The journal named the problem: a syntax error on line 12 of the enabled site1 file. I put the character back and reloaded. The site answered again after that.

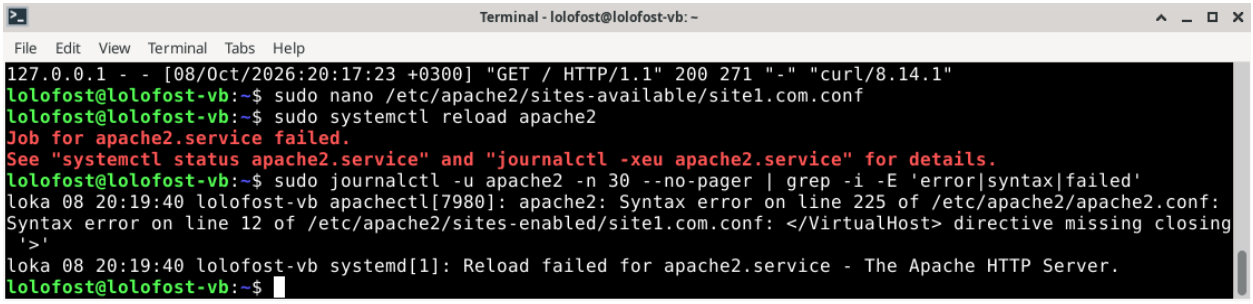


Image 11. The journal after the broken line.

site2.com

The second site uses the same address and a different name. I added site2.com and www.site2.com to /etc/hosts, put a different page in public-sites2, and wrote a second configuration file. a2ensite linked that file as well. sites-enabled now has the default site plus both of mine, and curl returns a different page for each name.

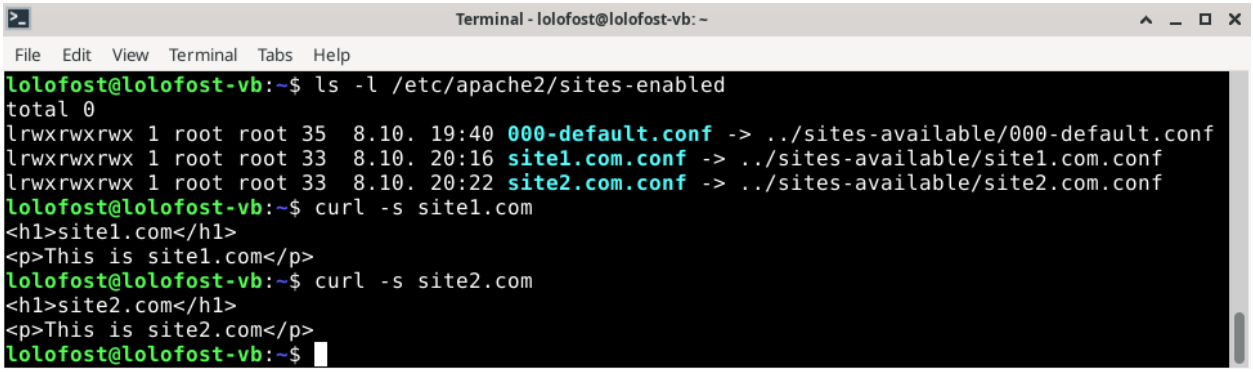


Image 12. Both sites enabled.

Notes

Two names can share one address. Apache chooses the site from the name in the request, so site1.com and site2.com both use 127.0.0.1 and still return different pages.